Yu-Jui Liang

2027 RDSS | Junior Engineer

liangyujui@gmail.com
+886 928 036 397
linkedin.com/in/yu-jui-liang-735a57316

EDA / VLSI CAD | Logic Synthesis | Physical Design / APR / STA | Digital IC / SoC / ASIC | Design Verification

EDUCATION

National Taiwan University *Sep. 2025 – Expected Jun. 2027*
M.S. Candidate, Data Science Master Degree Program
Research Advisors: Prof. Jie-Hong Roland Jiang (NTU); Dr. Hsiang-Yun Cheng (Academia Sinica)

National Chengchi University *Sep. 2021 – Jun. 2025*
B.S. in Computer Science; Double Major in Electronic Physics

EDA / IC RESEARCH & PROJECTS

Multi-Output Standard-Cell Mapping with MILP and Differentiable Optimization *Apr. 2026 – Present*
Master's Thesis */ Logic Synthesis / Technology Mapping / Multi-Output Cells / ASAP7*
- Extended a cut-based mapper with multi-output candidate discovery, Boolean matching, and MILP-based candidate selection; integrated the results into Liberty-aware differentiable technology mapping.
- Evaluated 30 arithmetic benchmarks, achieving **6% average ADP improvement** over the single-output mapping baseline while maintaining functional correctness.

ICCAD 2025 CAD Contest Problem C – STA-Driven Incremental Timing ECO Flow
STA / Timing ECO / Gate Sizing / VT Swapping / Buffer Insertion
- Developed an OpenROAD-based timing ECO flow applying critical-path gate sizing, VT swapping, and buffer insertion; explored candidate sequences with bounded best-first search, a branch list, and a global-best checkpoint.
- Evaluated five public benchmarks, achieving **16.7% median setup TNS** and **14.6% average setup WNS** improvements vs. seed DEFs, with approximately 8% lower power and 2% higher HPWL, verified by parasitic-aware STA/PPA and placement-legality checks.

Open-Source Mini-SoC and Macro-Based Physical Implementation *Dec. 2025 – Jan. 2026*
RTL-to-GDS / SRAM Macro Integration / Timing Closure / ASAP7
- Integrated a Wishbone-based Mini-SoC with SPI/I2C peripherals, an SRAM controller, and a 1-KiB SRAM hard macro; completed an ASAP7 RTL-to-GDS implementation with 11.1k instances at a 500-ps clock.
- Achieved post-route timing closure with **+34.8 ps setup slack**, **zero setup/hold violations**, and 8.94-mW power; **reduced die area by 16% and wirelength by 6%**, and extended the flow to an ORFS RV32I design with four 1-KiB SRAM macros, 18.4k instances, and 65% utilization.

ICCAD 2026 CAD Contest Problem C – Diffusion-Guided SoC Floorplanner
ML for EDA / Floorplanning / Macro Placement / HPWL Optimization
- Developed a diffusion-guided SoC floorplanner combining ML-based global initialization with combinatorial refinement for macro placement.
- Refined macro placements with constraint-aware simulated annealing to optimize HPWL and floorplan quality after diffusion-based initialization.

EXPERIENCE

AIMobile Co., Ltd. (Inventec Group) *Apr. 2025 – Jul. 2025*
Contract Software Engineer
- Contributed to software development, system integration, testing, and debugging in a corporate R&D environment.

Neopower Technologies Co., Ltd. *Jun. 2023 – Mar. 2025*
Contract Software Engineer
- Developed and maintained web applications, backend services, APIs, and data-processing workflows.

TECHNICAL SKILLS

Programming / HDL
C/C++, Python, Tcl, Bash, Verilog, SystemVerilog

EDA / IC Design
Logic synthesis, technology mapping, STA, physical design / APR, RTL-to-GDS, timing closure, timing ECO, PPA analysis

Tools / Platforms
PyTorch, CUDA, Linux, Docker